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Course Code: CSA0903
Course Name: Programming in JAVA

Class activity

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

RemoveDuplicates.java | Share | Run

```java
import java.util.ArrayList;
import java.util.Scanner;

public class RemoveDuplicates {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> list = new ArrayList<>();

        System.out.println("Enter  size :");
        int n=sc.nextInt();
        System.out.println("Enter  integers:");
        for (int i = 0; i < n; i++) {
            list.add(sc.nextInt());
        }

        for (int i = 0; i < list.size(); i++) {
            for (int j = i + 1; j < list.size(); j++) {
                if (list.get(i).equals(list.get(j))) {
                    list.remove(j);
                    j--;
                }
            }
        }

        System.out.println("After removing duplicates:");
        System.out.println(list);
    }
}
```

Output

```
Status: Accepted
Memory Used: 12300 byte | Execution Time: 0.177 sec

Enter  size :
Enter  integers:
After removing duplicates:
[1, 22, 23, 2, 0, 12, 3, 11]
```

Input

```
12
1 22 23 2 0 12 22 0 3 2 3 11
```

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

EmployeeList.java | Share | Run

```java
import java.util.LinkedList;
import java.util.Scanner;

public class EmployeeList {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        LinkedList<Integer> list = new LinkedList<>();

       // System.out.print("Enter number of employees: ");
        int n = sc.nextInt();

        //System.out.println("Enter Employee IDs:");
        for (int i = 0; i < n; i++) {
            list.add(sc.nextInt());
        }

        System.out.println("Forward Order:");
        for (int id : list)
            System.out.print(id + " ");

        System.out.println("\nReverse Order:");
        for (int i = list.size() - 1; i >= 0; i--)
            System.out.print(list.get(i) + " ");
    }
}
```

Output

```
Status: Accepted
Memory Used: 12720 byte | Execution Time: 0.203 sec

Forward Order:
101 102 103 104 105
Reverse Order:
105 104 103 102 101
```

Input

```
5

101
102
103
104
105
```

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

UniqueEmails.java

```java
import java.util.HashSet;
import java.util.Scanner;

public class UniqueEmails {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashSet<String> set = new HashSet<>();

        System.out.print("Enter number of emails: ");
        int n = sc.nextInt();
        sc.nextLine();

        System.out.println("Enter Email IDs:");
        for (int i = 0; i < n; i++) {
            set.add(sc.nextLine());
        }

        System.out.println("Unique Email IDs:");
        for (String email : set)
            System.out.println(email);

        System.out.println("Total Unique Emails: " + set.size());
    }
}
```

Output

```
Status: Accepted
Memory Used: 12640 byte | Execution Time: 0.202 sec

Enter number of emails: Enter Email IDs:
Unique Email IDs:

test@gmail.com
xyz@gmail.com
abc@gmail.com
Total Unique Emails: 4
```

Input

```
5

abc@gmail.com
xyz@gmail.com
abc@gmail.com
test@gmail.com
xyz@gmail.com
```

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

WordFrequency.java

```java
import java.util.HashMap;
import java.util.Scanner;

public class WordFrequency {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a paragraph:");
        String text = sc.nextLine();

        String[] words = text.split(" ");

        HashMap<String, Integer> map = new HashMap<>();

        for (String word : words) {
            if (map.containsKey(word))
                map.put(word, map.get(word) + 1);
            else
                map.put(word, 1);
        }

        System.out.println("Word Frequencies:");
        for (String word : map.keySet()) {
            System.out.println(word + " : " + map.get(word));
        }
    }
```

Output

```
Status: Accepted
Memory Used: 12956 byte | Execution Time: 0.208 sec

Enter a paragraph:
Word Frequencies:
java : 2
powerful : 1
is : 2
easy : 1
```

Input

```
java is easy java is powerful
```